

- ① An object is initially at 100°C is kept in a room, which is maintained at 20°C . If object cools to 40°C in 10 mins, find the temp of object after $1 \text{ hr.} = 60 \text{ min}$

Solution:- Newton's law of cooling
Let θ be the temp of object at time t

$$\frac{d\theta}{dt} \propto \text{temp. of object} - \text{temp of room}$$

$$\Rightarrow \frac{d\theta}{dt} \propto \theta - 20 \Rightarrow \frac{d\theta}{dt} = -k(\theta - 20)$$

$k > 0, \text{ constant}$

$$* \quad \frac{d\theta}{dt} = -k(\theta - 20)$$

$$\Rightarrow \int \frac{d\theta}{\theta - 20} = \int -k dt$$

$$\Rightarrow \log |\theta - 20| = -kt + C$$

When $t = 0$, $\theta = 100^{\circ} \Rightarrow C = \log 80^{\circ}$

$$\Rightarrow \log \left| \frac{\theta - 20}{80} \right| = -kt$$

When $t = 10$, $\theta = 40^{\circ}$

$$\Rightarrow \log \left| \frac{1}{4} \right| = -k \times 10 \Rightarrow \frac{1}{10} \log \left| \frac{1}{4} \right| = -k$$

$$* \quad \boxed{\log \left| \frac{\theta - 20}{80} \right| = \frac{t}{10} \log \left| \frac{1}{4} \right|}$$

$$\boxed{\quad \quad \quad | \quad 80 \quad | \quad \quad \quad}$$

Put $t=60$:- $\log \left| \frac{\theta - 20}{80} \right| = \frac{60}{10} \log \left| \frac{1}{4} \right| = \log \left| \frac{1}{4} \right|^6$

$$\Rightarrow \theta - 20 = \frac{1}{4^6} \times 80 \Rightarrow \frac{5}{4^4}$$

$$\Rightarrow \theta = \frac{5}{4^4} + 20$$

② Rate of growth of bacteria is proportional to amount present if population of 50 is present initially & population doubles every 4 hours, compute the population after 12 hrs.

Solution :- Let x be population of bacteria at time t

$$\Rightarrow \frac{dx}{dt} \propto x \Rightarrow \frac{dx}{dt} = kx, \quad k > 0$$

$$\Rightarrow \int \frac{dx}{x} = \int k dt \Rightarrow \boxed{\log x = kt + C}$$

$$\boxed{t=0, x=50} \Rightarrow \log 50 = C$$

$$* \log \left| \frac{x}{50} \right| = kt$$

$$\boxed{t=4, x=100} \quad \log 2 = 4k \Rightarrow k = \frac{1}{4} \log 2$$

$$* \log \left| \frac{x}{50} \right| = \frac{t}{4} \log 2$$

$$\therefore \quad t = 12 \quad \Rightarrow \log \left| \frac{x}{50} \right| = \frac{12}{4} \log 2 = \log 2^3$$

$$\text{When } t=12 \quad \Rightarrow \quad \log \left| \frac{x}{50} \right| = \frac{12}{4} \log 2 = \log 2^3$$

$$\Rightarrow \quad \frac{x}{50} = 2^3 \Rightarrow x = 2^3 \times 50 = 400.$$

③ Rate of decay of radioactive material is proportional to the amount present; if 10% decays in 200 hrs. Find the amount of material after 1000 hrs.

Solution:- Let x be amount of material present at time t

$$\frac{dx}{dt} \propto x \quad \Rightarrow \quad \frac{dx}{dt} = -kx$$

$$\Rightarrow \quad \log x = -kt + C$$

When $t=0$, let x_0 amount of material was present.

$$\Rightarrow \quad \log x_0 = C$$

$$\Rightarrow \quad \log \left| \frac{x}{x_0} \right| = -kt$$

$$t=200, \quad x = 0.9x_0 \quad \Rightarrow \quad \log \left| \frac{0.9x_0}{x_0} \right| = -k \times 200$$

$$\Rightarrow \quad \frac{1}{200} \log \left| \frac{9}{10} \right| = -k$$

$$* \quad \log \left| \frac{x}{x_0} \right| = \frac{t}{200} \log \left| \frac{9}{10} \right|$$

put $t = 1000$

$$\log \left| \frac{x}{x_0} \right| = \frac{1000}{200} \log \left| \frac{9}{10} \right|$$

$$\Rightarrow x = \left(\frac{9}{10} \right)^5 \cdot x_0$$

System of ODE

* Order n ODE in $\mathbb{R}$

= first order ODE in $\mathbb{R}^n$.

• Convert the following ODE into first order ODE in $\mathbb{R}^2$

$$2y'' - 5y' + y = 0, \quad y(3) = 6, \quad y'(3) = -1.$$

* independent variable t , $y' = \frac{dy}{dt}$, $y'' = \frac{d^2y}{dt^2}$

$$\rightarrow y'' = +\frac{5}{2}y' - \frac{1}{2}y$$

$$\circ \quad x_1(t) = y(t) \Rightarrow \boxed{x_1(3) = 6}$$

$$\circ \quad x_2(t) = y'(t) \Rightarrow \boxed{x_2(3) = -1}$$

$$x_1'(t) = y'(t) = x_2(t)$$

$$x_2'(t) = y''(t) = -\frac{1}{2}y + \frac{5}{2}y' = -\frac{1}{2}x_1 + \frac{5}{2}x_2$$

$$\begin{cases} x_1'(t) = x_2(t) \\ x_2'(t) = -\frac{1}{2}x_1(t) + \frac{5}{2}x_2(t) \end{cases}$$

$$X = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} \Rightarrow X' = \begin{pmatrix} x_1'(t) \\ x_2'(t) \end{pmatrix}$$

$$\begin{bmatrix} x_1'(t) \\ x_2'(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1/2 & 5/2 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

$$X' = AX \rightarrow \text{first order ODE in } \mathbb{R}^2$$

Initial conditions
 $x_1(3) = 6$
 $x_2(3) = 1$

* Write the following $y^{(4)}$ order differential equation as a system of first order linear differential eqn.

$$y^{(4)} + 3y'' - \sin(t)y' + 8y = t^2$$

$$y(0) = 1, \quad y'(0) = 2, \quad y''(0) = 3, \quad y'''(0) = 4$$

Solution:-

$$\begin{aligned} x_1(t) &= y(t) & x_2(t) &= y'(t) \\ x_3(t) &= y''(t) & x_4(t) &= y'''(t) \end{aligned}$$

$$x_1'(t) = y'(t) = x_2(t)$$

$$x_2'(t) = y''(t) = x_3(t)$$

$$x_3'(t) = y'''(t) = x_4(t)$$

$$\dots \dots \dots y^{(4)}(t) = -3y'' + \sin t y' - 8y + t^2$$

$$\begin{aligned}
 x_3'(t) &= y'''(t) = x_4(t) \\
 x_4'(t) &= y^{(4)}(t) = -3y'' + \sin t y' - 8y + t^2 \\
 &= -3x_3(t) + \sin t x_2(t) - 8x_1(t) + t^2
 \end{aligned}$$

$$\begin{aligned}
 x_1'(t) &= x_2(t) \\
 x_2'(t) &= x_3(t) \\
 x_3'(t) &= x_4(t) \\
 x_4'(t) &= -3x_3(t) + \sin t x_2(t) - 8x_1(t) + t^2
 \end{aligned}$$

$$\begin{bmatrix} x_1' \\ x_2' \\ x_3' \\ x_4' \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -8 & \sin t & -3 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ t^2 \end{bmatrix}$$

Initial conditions:- $x_1(0)=1, x_2(0)=2, x_3(0)=3, x_4(0)=4$

$X' = AX + g(t)$ $\rightarrow$ Non hom Eqn in $\mathbb{R}^n$

• If $g(t)=0$, then it is called Homogeneous Eqn in $\mathbb{R}^n$.

• How to solve $\boxed{X' = AX} \rightarrow \mathbb{R}^n$

• in $\mathbb{R}$ $x' = ax \Rightarrow \frac{dx}{dt} = ax$

$$\Rightarrow \int \frac{dx}{x} = \int a dt$$

$$\Rightarrow \log x = at + \log C$$

$$\Rightarrow x = Ce^{at}$$

$$\rightarrow \log x = \dots$$

$$\Rightarrow x = C e^{at}$$

Trial solution of $x' = Ax$ is $v e^{at}$

$$x = v e^{at}$$

$$x' = a v e^{at}$$

$$a v e^{at} = A v e^{at}$$

$$\Rightarrow (A v - a v) e^{at} = 0$$

$$x' = Ax$$

$v e^{at}$ $\rightarrow$ eigen value
 $\downarrow$
 eigen vector of A

$$\Rightarrow A v - a v = 0 \Rightarrow \boxed{A v = a v}$$

$$\Rightarrow (A - a) v = 0$$

$v \rightarrow$ eigen vector of A corresponding to eigen value a